MECHANICS OF STEM CELL DIFFERENTIATION

Hamsini Suresh[1]*, Andrea Vigliotti[1,2], Siamak Shishvan[1] & Vikram Deshpande[1]

[1] Department of Engineering, University of Cambridge, Cambridge CB2 1PZ, UK

[2] Innovative Materials Laboratory, Italian Aerospace Research Centre, 81043 Capua, Italy

Cells can 'feel' their environment

Cells exist in a fluctuating equilibrium with their surroundings. The external environment influences these fluctuations which in turn control cell differentiation. **Fig. 1** shows one such example.

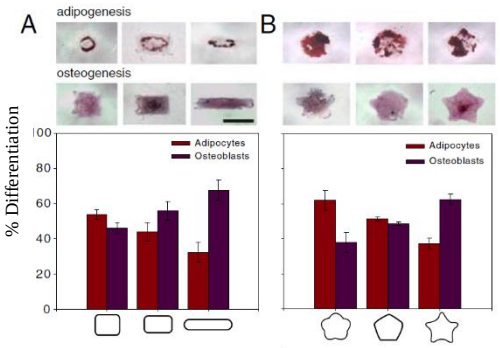


Fig. 1 Cell shape governs stem cell differentiation. [Kilian K., et al, PNAS 107, 2010]

Predicting these fluctuations is key to understanding the response of cells to mechanical, topological and chemical cues.

Model Overview: A Statistical Mechanics Approach

Consider the thermodynamic system illustrated in **Fig. 2**.

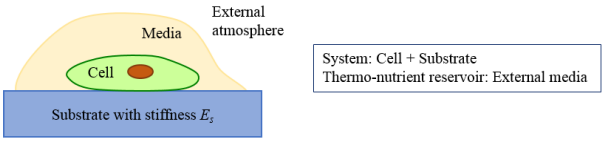


Fig. 2 Cell + substrate system surrounded by nutrient pool and external atmosphere.

For this system, define a spread micro-state as a specification of connection of material points on the cell membrane to those on the substrate.

↓

Fluctuations of the cell correspond to the fluctuations of the system between different spread micro-states.

↓

At every micro-state, the cell is at mechanical and chemical equilibrium.

↓

Minimising the free-energy of the system for a given micro-state gives the probability of a micro-state i with equilibrium free-energy G_i as:

$$P_i \propto e^{-\beta G_i}$$

(β is given by cellular homeostasis requirement)

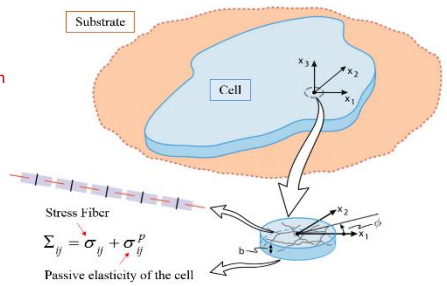
↓

Monte-Carlo simulations to explore this enormous phase-space (typically explore 10^6 or more equilibrium configurations) and get probability distributions on any system variable or observable.

Simple model for free energy of a micro-state

For a given micro-state i, $G_i^{tot} = G_i^{cyto} + G_i^{elas}$

Fig. 3 Simplest model for G_i^{tot} of a micro-state has only two components: actin/myosin cytoskeleton (stress-fibers) and the passive elasticity of the cell. [Vigliotti A., et al, BMMB 2015]



Cell on 2D square patches of different sizes

Experiment: When cells are confined to square patches of different sizes, cells spread more and increasingly differentiate into osteoblasts on larger patches and vice versa.

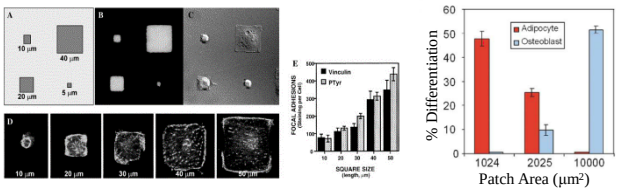


Fig. 4 On a larger patch, **a** cells form more focal adhesions [Chen C., et al, BBRC 307, 2003] and **b** more cells prefer to form osteoblasts [McBeath R., et al, Dev. Cell 6, 2004].

Differentiation analysis: Information for the fluctuations is encoded in the probability distribution of the cell's cytoskeletal free energy distribution (**Fig. 5a**).

Assuming that a cell differentiates into a particular type when its average cytoskeletal free-energy attains a specific value, using the Central Limit Theorem, we can estimate the probability that a cell differentiate into an osteoblast or an adipocyte:

$$P_{osteo/adipo} = \frac{1}{\bar{\sigma}\sqrt{2\mu}} e^{\frac{-[G_{osteo/adipo}-\mu]^2}{2\bar{\sigma}^2}}$$

Where N_c is the number of states used to estimate the differentiation probability, μ and σ are the mean and variance of the underlying distribution, with $\bar{\sigma} = \frac{\sigma}{\sqrt{N_c}}$ (**Fig. 5b**).

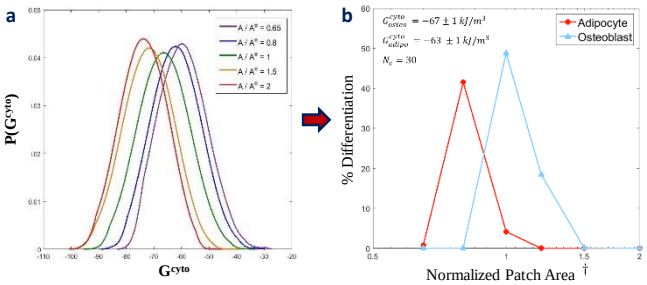


Fig. 5a G^{cyto} distributions for cell on 5 different patch sizes and **b** result of the differentiation analysis on 5 different patch sizes (G_{adipo}^{cyto} and G_{osteo}^{cyto} (in inset) are chosen to reflect the experimental observation).

† A/ A^e ≡ patch area is normalized w.r.t area of cell under zero strain

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